Talk to Transformers: An Empirical Study of Device Communications for the FREEDM System

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***Abstract*—The smart grid is a brand new power system that incorporates power infrastructures with information technologies. In the smart grid, power devices are interconnected to facilitate various intelligent applications via ubiquitous communications between devices. Currently, communications between equipments are mainly enabled by a centralized supervisory control and data acquisition (SCADA) system. An interesting question is *whether such SCADA systems and corresponding protocols can be readily used in the smart grid to satisfy performance requirements of intelligent applications*. To address these issues, in this paper, we establish a monitoring system for a Solid State Transformer (SST) in a micro smart grid, *Green Hub*, to measure achieved delay performances in a really deployed system. Based on our results, we verify that the widely-used SCADA system shows satisfactory performances to meet requirements defined in Green Hub for device monitoring and control. Besides the performance evaluation, we also intend to share our implementation experiences and highlight several significant remarks for the future design of the communication architecture in the smart grid.**

I. Introduction

The smart grid is an emerging cyber-physical system that incorporates power infrastructures with information technologies [1] to enable an efficient power management paradigm. The most salient feature in the smart grid is that various power equipments are interconnected via a communication infrastructure. Thereby, miscellaneous device information can be easily accessed by other devices or system operators in a more flexible and ubiquitous way. Moreover, various advanced automation applications, such as relay protection [2] and demand response [3], will also benefit from such ubiquitous device communications to make the power infrastructure more efficient and intelligent. In this sense, an appropriate communication architecture for device communications is of great importance in the smart grid.

Currently, the communication in the power system mainly depends on the centralized supervisory control and data acquisition (SCADA) system, in which equipment information is presented in unified manners following several widely-adopted SCADA protocols, such as distributed network protocol 3.0 (DNP3) [4] and Modbus [5]. Although the current SCADA system played a crucial role in the past decades of years for well-maintained power systems, it is often challenged for its out-of-date design, that is, SCADA system components, from communication protocols to underlying physical connections, were originally designed over serial links, such as RS-232 and RS-485. Towards the smart grid, where TCP/IP based internet technologies are extensively proposed, the critics questions *whether current SCADA systems or protocols can satisfy the performance requirements of applications in the smart grid.* Furthermore, there indeed lacks literatures to share implementation experiences and insights on *how to design and establish communication architectures with efficient device communications for the smart grid applications*.

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Motivated by these questions, we establish a communication architecture for a micro smart grid, *Green Hub*, in the Future Renewable Electric Energy Delivery and Management (FREEDM) systems center [6]. More specifically, we implement a Solid State Transformer (SST) [7] monitoring system used in Green Hub to achieve bidirectional communications between a control center and a real SST, which makes data exchange like a talking or conversation between devices. Through such a system integration for the “device talking”, we intend to highlight several important remarks as the learned lessons on the design and implementation of communication infrastructures in the smart grid for various applications.

The remainder of this paper is organized as follows. In Section II, we briefly introduce the system background, including the physical and communication architecture of the Green Hub system, and timing requirements of different applications. In Section III, we present the system setups of the SST monitoring system. In Section IV, we illustrate our experimental results and discuss our findings. Then, the learned lessons are summarized in Section V. Finally, we conclude in Section VI.

II. Green Hub - A Micro Smart Grid

In this section, we present the physical and communication architecture of Green Hub. And then, we introduce timing requirements defined in Green Hub for different applications.

A. Background of Green Hub

The one megawatt Green Hub system [8], [9] is a power electronics based power system in the FREEDM systems center [10], [6]. It is established to demonstrate salient features and capabilities of the notional FREEDM system on renewable energy generation, distribution, storage and management. In Green Hub, a variety of distributed renewable energy generators, such as solar panels and wind turbines, are envisioned

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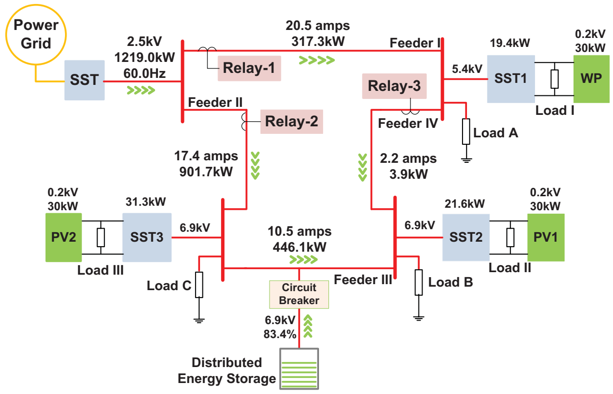

Fig. 1. The physical architecture of Green Hub.

to be integrated with diversified power equipments to form a small-scale power distribution network.

As shown in Fig. 1, there is a turbine-based wind power (WP) subsystem, and two solar-array based photovoltaic (PV) subsystems PV1 and PV2 in Green Hub, which are energy supplies for Loads I, II and III, respectively. Power generated by each renewable energy subsystem can be also delivered via SSTs to Loads A, B and C to alleviate the burden of the conventional power grid. In addition, extra energy can be stored in the distributed energy storage (DES) system for the future use. A circuit breaker (CB) is installed in the DES system to control its connectivity to Green Hub. Additionally, to ensure the reliability of Green Hub with flexible power flows, protective relays (Relays 1, 2, and 3 in Fig. 1), which are electrically operated switches, are deployed to handle potential faults or anomalies on power feeders.

B. Communication Architecture in Green Hub

As we can see, Green Hub is a small-scale distributed energy generation and delivery system. Yet, it consists of critical smart grid components, including WP systems, PV systems, SSTs, the DES system and relays. To facilitate efficient communication between these components, a 100Mbps Ethernet based communication infrastructure is deployed in Green Hub. As shown in Fig. 2, each power equipment is connected to an intelligent electronic device (IED), which is a microprocessor-based controller serving as the interface between a power device and the communication network. For example, PV systems and SSTs are connected to PV and SST controllers, respectively. All IEDs are interconnected via an Ethernet switch to form a single-hop communication network. Also, the Green Hub control center is connected to this network to support centralized management.

With such a single-hop communication network, the next essential component for the notional "device talking" is a full-fledged communication protocol, which presents data of different devices in a unified manner. In Green Hub, we employ the DNP3 over TCP/IP protocol stack as such a protocol deployed on all devices for message delivery. The reason we adopt the DNP3 over TCP/IP architecture is that

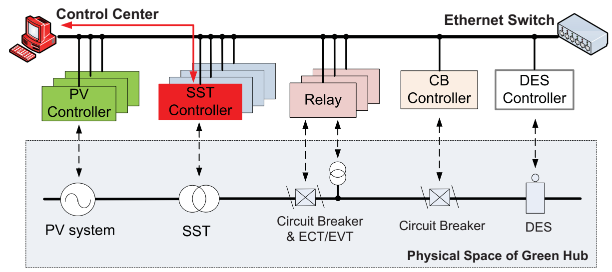

Fig. 2. The communication infrastructure in Green Hub.

TABLE I
REQUIREMENTS OF MESSAGE DELIVERY DELAY IN GREEN HUB.

Type	Delivery delay	Applications
Protection	3 ~ 16 ms	Trip, Closing, Reclosing
Monitoring and Control	16 ~ 100 ms	State reporting and control
Low-speed	≥ 100 ms	file transferring

it is the most popular one used in the current power systems [11], [12], [13], which may make our evaluation results close to practical situations in the power system.

Through the Ethernet connection and the communication protocol, multiple "device talking" pairs form in Fig. 2, such as Control Center/PV controller, SST Controller/Relay, and so on. Among pairs, the most important one is the Control Center/SST controller since the SST serves as an envisioned "Energy Router" [6] to "route" energy for different locations under instructions of the control center. In this manner, communication with the SSTs is the primary task for the implementation of Green Hub. Thus, in the remaining sections, we focus more on a DNP3-based SST monitoring system from the control center to the SST controller, marked by a red line in Fig. 2. We expect to present some implementation experiences and corresponding performance observations via such a monitoring system. As an implementation example of device communications, similar method can also be readily deployed in the remaining "device talking" pairs.

C. Communication Requirements in Green Hub

As a critical part of power management systems, communications between devices must meet some performance requirements in Green Hub to ensure the system in a normal state. Different from conventional networks, applications in Green Hub concern more about the delay performance since many tasks of data collections and controls are required to be finished in assigned time periods. Otherwise, out-of-date messages could be invalid and result in potential system failures. For example, when a short circuit happens on a feeder, the SST should deliver the overcurrent report immediately to the control center to trigger corresponding system responses for timely fault clearances. Table. I illustrates timing requirements for applications in Green Hub [14]. Among such, the allowable delay range for the device monitoring and control is from 16ms to 100ms. As a practical monitoring case, our SST monitoring system needs to meet such a performance requirement.

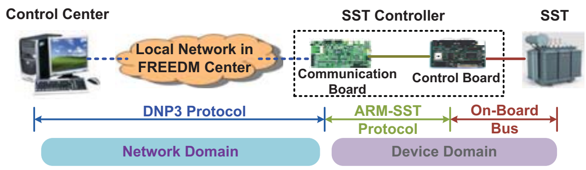

Fig. 3. SST Monitoring System.

III. SST Monitoring System Setup

As mentioned before, in this section, we will skip the setup description of the entire communication architecture in Green Hub, and mainly focus on the implementation details of the SST monitoring system.

A. System Architecture of the SST Monitoring System

The objective of the SST monitoring system is two-fold: (1) the SST can "speak" to the control center that implies the data acquisition from the SST to the control center; (2) the control center can also "speak" to the SST, which means distributions of control commands from the control center to the SST. To achieve such bidirectional communications between the SST and the control center, we design a system architecture as shown in Fig. 3.

The SST monitoring system is composed of two domains, network domain and device domain. In the network domain, a control center is connected to the SST controller via a Local Area Network (LAN) established by a TRENDnet TE100-S8P 10/100 Mbps Ethernet Switcher in the FREEDM center. Both the control center and the SST controller are equipped with a DNP3 protocol stack provided by the Green Energy Corp. [15]. Thereby, the collected data and dispatched control commands can be delivered via DNP3 packets between devices. In terms of the device domain, it features two components, a SST device and a SST controller. The SST device is in charge of the fundamental functions of a transformer, such as voltage step-down or step-up, harmonic filtering and voltage sag correction [16]. The SST controller is the interface between the device and the outside central controller, which is used to forward sampled data from the device to the control center as well as the commands from the control center to the device.

As mentioned, the SST controller plays dual roles in the monitoring system, one is the "courier" to communicate data or commands between the control center and the device; the other is the "controller", that means, the command will be translated into signals here for device control. In this case, the SST controller needs both communication and control capabilities. To achieve such a dual-role controller, we adopt a dual-board framework to separate the control and communication functions into two different boards, a DSP board for control and an ARM board for communication. The reason of such a dual-board framework lies in the complementary features between DSP and ARM. The former one is good at the high speed digital signal processing, but has little communication capability; whereas the latter one can provide communication interfaces. Thus, the two boards are integrated together as the SST controller with both control and communication capabilities. More specifically, we employ a *DSP C6713* as the control board, running *μCOS II* real time operation system, whose main function is to provide a powerful calculation capability for the processing of current/voltage sampling from the SST. In the meanwhile, a *TS-7250* ARM board is deployed as the communication board.

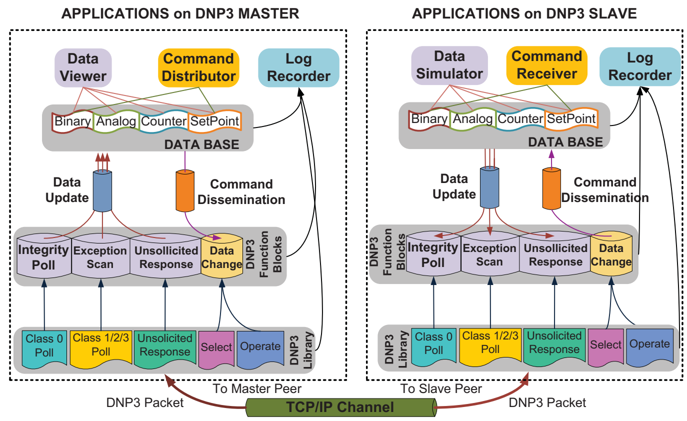

Fig. 4. Applications Setup over DNP3.

The two boards are connected with each other through a RS-232 based serial link. To ensure a unified data representation between the two boards, we develop an interface protocol, called as ARM-SST protocol, to be a protocol translator between the DNP3 packet and the data format used in the control board. For example, a command from the control center is enveloped in a DNP3 packet, which will be translated to a data format that can be understood by the control board to generate corresponding control signals.

B. Applications Setup

With such a hardware architecture described before, the next step is to establish monitoring related applications serving for data acquisition and command distribution. In the following description, we illustrate the application setups, including applications in network domain and those in device domain.

1) Network Applications Setup: Since DNP3 over TCP/IP is employed as the underlying communication protocol, the design principal of network applications is to leverage basic DNP3 functions for the SST monitoring. We then introduce the network applications setup in a top-down manner.

Fig. 4 depicts the application design in the network domain. Since DNP3 commonly works in a master/slave mode, the applications setup is correspondingly divided into two parts for the master and the slave. We envision three basic tasks on the master, including data query, command distribution and log collection, which are mapped into three modules, data viewer, command distributor and log recorder respectively.

- *Data Viewer* is to exhibit real-time data values on the SST. The data can be categorized into four types, binary, analog, counter and set point, regarding the DNP3 specification [4]. Each data present a determined parameter of the SST, such as the current, voltage or temperature.

- *Command Distributor* serves for the command generation that is presented by a series of modified binary or set point data as per the DNP3 specification.
- *Log Recorder* is a log file collector that is embedded in the whole application system to record data changes or events occurred in the system for the off-line analysis.

To support data query and modifications in *Data Viewer* and *Command Distributor*, a data base is programmed to store data from the device or issued by *Command Distributor*. Two data channels are furnished under the data base to retrieve data updates from the DNP3 function blocks, or dispatch commands to function blocks. The DNP3 function blocks aggregate four DNP3 functions used in our applications, including integrity poll, exception scan, unsolicited response and data change. These four functions are actually mapped to the action items in DNP3 library. For example, the integrity poll is executed by "Class 0 Poll"; while the data change involves two continuous actions, "Select" and "Operate".

As DNP3 is overlayed over TCP/IP in the Green Hub, the generated DNP3 packet will be finally fed to the slave via a TCP/IP channel. On the slave side, the application design is the same with that on the master. Note that, since the development of the SST is still in progress, we here use a data simulator to temporarily replace the real SST as a data source.

2) Device Applications Setup: The applications on the device domain are achieved by the earlier mentioned ARM-SST protocol. All operations issued from the control center are translated to various signals that can be understood by the SST. These signals are further interpreted to be diverse states of high-speed switches on the SST. In this way, the expected voltage or current output can be achieved [7].

Therefore, we have established a SST monitoring system prototype in Green Hub to "talk" with the SST through several applications. In the following section, we deploy several experiments to verify its performances in real control operations.

IV. Performance Measurements and Analysis

In this section, we intend to answer the open question *whether current SCADA systems or protocols can be qualified for the device communication in the smart grid* via a series of experiments. Furthermore, we summarize performance results and learned lessons to provide some insights for the future communication design in the smart grid.

As shown in Table. I, applications' delay performances in Green Hub need to satisfy various timing requirements. For the SST monitoring system, since its main task is to supervise devices and clear possible equipment anomalies, the delays for the fault finding and clearing should be primary metrics for performance evaluations. To investigate the fault finding and clearing delay, we start from a completed message delivery process among system components, including the control center, the communication board and the control board. In other words, a diagram about message delivery between system components is significant for the delay analysis. In the following part, we illustrate a detailed delay analysis according to the message delivery diagram in two DNP3 operation modes, event-driven mode and non-event-driven mode.

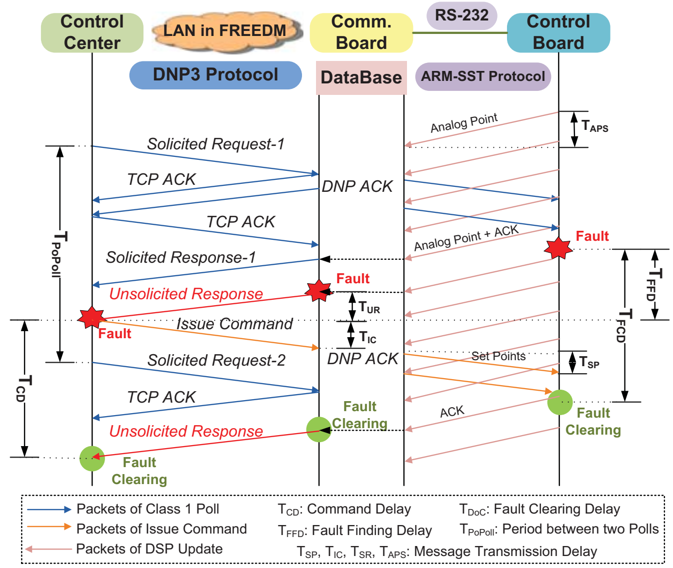

Fig. 5. Message Sequence for Fault Clearing in DNP3 Event-Driven Mode.

A. Message Diagram of DNP3 Event-Driven Mode

In accordance to the DNP3 specifications, DNP3-enabled devices are able to be configured to work in two different modes, event-driven mode and non-event-driven mode depending on the system requirements [4], [17]. In DNP3 event-driven mode, a device can spontaneously initiate message transmissions to report its own data change or event occurrence without a specific request.

As shown in Fig. 5, when a fault happens, it will be firstly observed by the control board through one or multiple out-of-range "analog points" on the device, such as voltage or current. Then, the fault information is delivered to the communication board by periodical samplings issued by the control board. Since the communication board is configured on the DNP3 event-driven mode, the arrival fault message will be immediately send back to the control center via a DNP3 Unsolicited Response message. Based on the fault message, the control center issues several "set points" as the corresponding commands to clear the fault by changing ON/OFF states of high speed switches on the SST. When the fault is cleared, the out-of-range "analog points" will fall back into the normal range that will trigger another Unsolicited Response packet to inform the control center such a new running state. In this process, we concern more about three delays, including Fault Finding Delay (FFD), Fault Clearing Delay (FCD) and Command Delay (CD).

With the message diagram shown in Fig. 5, we summarize the calculations of the three critical delays in Table. II. We can see that FFD is mainly determined by the transmission delay of "analog point" sampling T_{APS} and the transmission delay

TABLE II
DELAY CALCULATIONS FOR FAULT FINDING AND CLEARING IN EVENT-DRIVEN MODE.

Delays	Delay Expression
Fault Finding	$T_{FFD} = T_{APS} + T_{UR}$
Fault Clearing	$T_{FCD} = T_{FFD} + T_{IC} + nT_{SP}$
Command Control	$T_{CD} = (T_{FCD} - T_{FFD}) + T_{APS} + T_{UR}$

TABLE III
AVERAGE DELAY IN DNP3 EVENT-DRIVEN MODE.

Delay	Delay Components		Average Delay (ms)
	Comp.	Comp. Delay (ms)	
FFD	T_{APS}	3.79	11.17
	T_{UR}	7.48	
FCD	T_{FFD}	11.17	30.31
	T_{IC}	0.26	
	T_{SP}	18.97	
CD	T_{FCD}	30.31	30.42
	T_{FFD}	11.17	
	T_{APS}	3.79	
	T_{UR}	7.48	

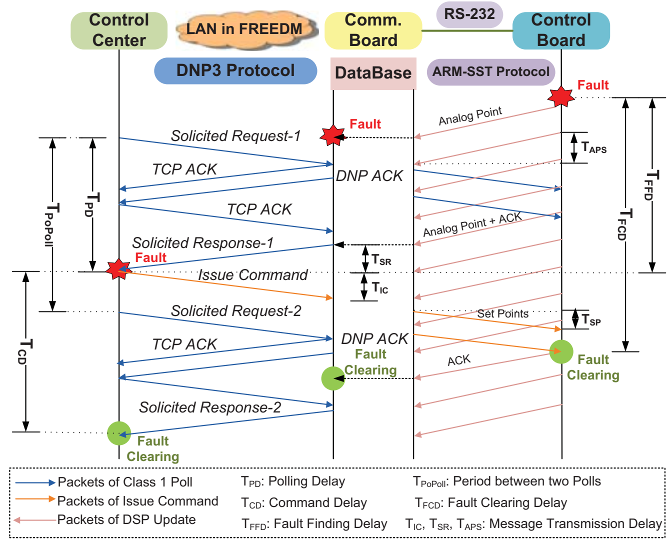

Fig. 6. Message Sequence for Fault Clearing in DNP3 Non-Event-Driven Mode.

of Unsolicited Response message T_{UR} [1]; while FCD depends on FFD, the command transmission delay T_{IC} and the delay for those "set points" accepted by the control center nT_{SP}. The command delay relies on both FCD and FFD, and also is affected by transmission delays of sampling and unsolicited response message.

B. Performance Results of DNP3 Event-Driven Mode

As per delay calculations listed in Table. II, we evaluate FFD, FCD and CD shown in Table. III. We can see that, if the SST fault occurs at time 0, then it is observed by the control center at around 11ms, and cleared at 30ms. It takes 30ms for operators to verify the fault clearing through the updated Unsolicited Response message after the issued command.

Compared with the timing requirements shown in Table. I, the Fault Finding Delay satisfies the timing requirements for monitoring. In terms of FCD and CD, they also show sound performances that can be accepted in an equipment monitoring and control scenarios of Green Hub. Although current delay performances are satisfactory, there still exists many options to optimize delay performances for more time-critical applications, such as rearranging the task scheduling on the control board and improving protocol efficiency, which are planned as the future work.

C. Message Diagram of DNP3 Non-Event-Driven Mode

Besides the Event-Driven mode, DNP3-enabled devices can also be configured in a Non-Event-Driven mode [4], [17]. Different from the Event-Driven mode, a DNP3 Non-Event-Driven device is not able to initiate a data response unless it receives a data request in advance. In this sense, a master device needs to firstly issue a data request, called as polling in DNP3, and then the slave device responses the request with the updating messages. The message delivery diagram is shown in Fig. 6.

In DNP3 Non-Event-Driven mode, a SST fault report is still delivered to the communication board by the periodical sampling from the control board. However, although the fault has been identified in the communication board, the device can not immediately send the fault back to the control center. Adversely, it needs to hold the fault report for the next arrived polling. There are several polling types defined in DNP3 [4]. In this paper, we take the Class 1 poll as the example since Class 1 poll is defined with the highest priority and only reports the data that has changed [17]. In practice, a Solicited Request message of Class 1 poll triggers a Solicited Response message issued to carry the fault report. As for the fault clearing process, it is almost the same as that in the Event-Driven mode. Only difference lies in that the fault clearing report is delivered by another Solicited Response, not the Unsolicited Response.

Similarly, we take three delays as performance metrics, including the fault finding delay, the fault clearing delay and the command delay. The delay calculations are presented in Table. IV. Different from the delay in the Event-Driven mode, delays in the Non-Event-Driven mode are not fixed values, but fall in a delay range since the delays are determined by the interval between the fault arrival and the first Solicited Response. In the best case, the fault report arrives right earlier than the Solicited Response. Then, the fault can be send back without any delay. In the worst case, the fault report arrives right after the previous Solicited Response in order that it has to wait for the next one. Therefore, the three delays significantly depend on the frequency of Class 1 poll.

To ensure a shorter delay, the frequency of Class 1 polling should be configured as high as possible so that the waiting time of the fault report can be reduced significantly. Ideally, the

[1] Note that we assume that the sampling rate on the control board is high enough in order that the fault can be issued right after its occurrence.

[2] $T_{Min} = (T_{FCD} - T_{FFD}) + T_{APS} + T_{SR}$.

TABLE IV
DELAY RANGES FOR FAULT FINDING AND CLEARING IN NON-EVENT-DRIVEN MODE.

Functions	Delay Ranges (ms)
Fault Finding	$T_{APS} + T_{SR} \sim T_{PoPoll} + T_{APS} + T_{SR}$
Fault Clearing	$T_{FFD} + T_{IC} + nT_{SP}$
Command Control	$T_{Min} \sim T_{Min} + T_{PoPoll}$ [2]

polling period is short enough that it can be ignored. Then, results of the three delays will approach to those in DNP3 Event-Driven mode shown in Table. III. On the other side, a higher polling frequency implies a heavier network traffic on the network, which may lead more congestions or interferences in turn to prolong achieved delays. Therefore, a fine-grained trade-off is in essence between the delay performance and network traffic load.

V. Lesson Learned

Through illustrated performance results, we can answer the question referred in the beginning of this paper, *"whether current systems or protocols can be qualified for the device communication in the smart grid"*. The dominant DNP3 based SCADA system can be used in the smart grid for the device monitoring and control, but need a meticulous implementation and configuration. We then summarize the learned lessons from our implementation experiences.

- Since many applications in the smart grid are delay-oriented with rigorous timing requirements, delay should be the primary performance concern for the communication infrastructure. In many cases, the "delay ingredients" are compound, involving network transmission delay, on-board processing delay and even propagation delays between components inside the device, like the delay between the communication board and the control board in our SST controller.
- Due to the complicated "delay ingredients", a careful optimization is much more crucial to reduce the total delay by optimizing every timing consuming part of every system component. For example, in our implementation of the SST controller code, although the delay performance satisfies the timing requirement, there still exists possibilities to achieve lower latency through various optimizations on system architecture or protocol.
- Besides the monitoring and control, there are other applications in the smart grid that involve device communications with more rigorous delay requirements, such as relay protection. In a relay protection, a relay needs to "talk" with other devices in several milliseconds. In such a case, a simple protocol architecture is better, while the DNP3-based SST monitoring system is not suitable any more since its architecture is so complex that it induces many extra delays. For example, the dual-board framework introduces more latencies on communication between boards.
- To achieve a lower delay in the smart grid, a fine-grained system configuration is another important factor, which should be deduced from a comprehensive system understanding. For example, in DNP3 Non-Event-Driven mode, in spite of the fact that a shorter period of polling can save the waiting time of the fault report, it may dramatically increase the network traffic. The congestions or interferences from the increased traffic in turn depress the delay performance. Therefore, an optimized system configuration is critical for system performances.

VI. Conclusion

In this paper, we established a SST monitoring system in a micro smart grid, Green Hub, to provide an experimental study for the delay performance of SCADA systems. Our experimental results revealed that the DNP3-based SCADA system can satisfy the performance requirements of equipment monitoring in the smart grid. Also, we illustrated the implementation details, as well as our implementation experiences upon deployments of the communication architecture in the smart grid. Our future work is to polish the delay performance of the system via optimizations and to extend the current system to incorporate more devices.

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